

POINTE-AU-PERE, QC, Canada

WMO: 715540

Lat: 48.5139N

Lon: 68.4683W

Elev: 5

StdP: 101.27

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.2	-26.8	0.3	-21.3	-24.1	0.4	-18.7	15.7	-7.8	14.3	-8.6	6.2	290	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	23.1	18.6	21.5	17.7	20.2	16.7	19.6	22.3	18.3	20.9	17.2	19.6	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.3	21.5	17.1	12.2	20.2	16.0	11.4	18.9	56.2	22.4	52.1	21.0	48.4	19.7	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.8	11.2	9.9	DB	-25.1	28.5	3.2	2.4	-27.4	30.2	-29.3	31.6	-31.1	32.9	-33.5	34.6	
			WB	-25.4	22.6	3.1	1.2	-27.6	23.5	-29.4	24.2	-31.1	24.9	-33.4	25.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-9.8	-8.8	-4.1	2.3	8.2	12.8	15.8	15.2	11.6	6.3	0.9	-5.4
	DBStd	9.72	5.92	5.31	4.90	3.33	3.26	3.11	2.41	2.42	3.08	3.35	4.14	5.00
	HDD10.0	2787	614	526	438	231	75	8	0	0	17	126	274	478
	HDD18.3	5312	873	759	696	480	313	168	85	102	202	374	523	736
	CDD10.0	529	0	0	0	1	21	92	179	161	66	9	1	0
	CDD18.3	12	0	0	0	0	0	2	6	4	1	0	0	0
	CDH23.3	50	0	0	0	0	2	9	22	16	2	0	0	0
	CDH26.7	7	0	0	0	0	0	1	3	3	0	0	0	0
Wind	WSAvg	4.5	5.5	5.1	5.1	4.6	3.9	3.5	3.4	3.4	3.8	4.6	5.3	5.5
Precipitation	PrecAvg	947	57	62	72	68	81	88	88	83	91	92	77	96
	PrecMax	1109	101	113	134	128	143	172	153	157	255	143	156	134
	PrecMin	696	22	28	42	34	45	33	17	26	22	32	26	49
	PrecStd	109	23	24	28	29	29	36	35	36	51	33	27	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	4.7	8.3	14.3	21.5	24.5	25.7	25.2	22.9	19.2	16.2	7.1
		MCWB	5.3	3.3	5.8	9.7	15.1	18.6	21.0	19.9	18.6	14.9	13.4	5.7
	2%	DB	3.1	1.7	5.2	10.4	18.0	21.8	23.1	22.3	20.1	15.6	11.1	4.2
		MCWB	2.1	0.6	3.0	6.3	12.9	17.0	19.1	18.4	16.7	12.6	8.6	3.2
	5%	DB	0.8	0.4	3.6	8.3	15.5	19.7	21.6	20.8	18.1	13.0	7.8	2.5
		MCWB	-0.1	-0.5	1.8	5.1	11.2	15.6	18.0	17.4	15.2	10.4	6.0	1.4
	10%	DB	-1.4	-0.9	2.3	6.7	13.4	18.0	20.2	19.4	16.5	11.0	5.9	1.2
		MCWB	-2.4	-1.8	0.7	4.1	9.9	14.5	16.9	16.5	13.9	8.9	4.4	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	3.6	6.1	10.0	16.5	20.3	21.8	20.9	19.3	15.5	14.5	6.1
		MCDB	6.4	4.6	7.9	12.6	20.4	23.7	24.8	23.8	22.2	18.1	17.5	7.2
	2%	WB	2.1	0.8	3.4	7.0	13.8	17.7	19.9	19.1	17.2	12.7	9.0	3.3
		MCDB	2.9	1.5	4.7	9.6	16.9	20.8	22.4	21.5	19.5	15.1	10.9	4.1
	5%	WB	0.0	-0.4	2.0	5.5	11.7	16.2	18.5	17.9	15.5	10.8	6.2	1.5
		MCDB	0.9	0.3	3.4	7.9	14.9	19.1	21.0	20.3	17.8	12.7	7.6	2.3
	10%	WB	-2.3	-1.7	0.8	4.3	10.2	14.9	17.3	16.8	14.1	9.1	4.6	0.3
		MCDB	-1.4	-0.9	2.0	6.2	13.0	17.5	19.8	19.0	16.2	10.7	5.7	1.1
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.0	6.7	6.2	7.5	8.1	8.2	7.9	7.3	5.8	5.0	5.6
		MCDBR	8.4	6.9	7.3	9.3	11.5	10.7	10.2	9.8	9.8	9.2	7.8	6.8
		MCWBR	8.3	6.5	6.1	6.7	7.5	7.1	7.1	6.7	7.3	7.0	6.6	6.6
	5% WB	MCDBR	8.3	6.6	6.9	8.7	10.7	10.2	10.0	9.4	9.5	8.8	7.4	6.8
		MCWBR	8.2	6.4	6.1	6.6	7.6	7.4	7.5	7.1	7.5	7.0	6.5	6.7
Clear-Sky Solar Irradiance	taub	0.264	0.273	0.303	0.357	0.367	0.389	0.413	0.393	0.367	0.330	0.299	0.271	
	taud	2.346	2.315	2.292	2.297	2.347	2.351	2.317	2.374	2.428	2.513	2.460	2.379	
	Ebn at Noon	813	890	913	892	894	873	844	847	835	817	766	755	
	Edh at Noon	68	88	108	120	119	120	122	111	95	74	62	59	
All-Sky Solar Radiation	RadAvg	1.26	2.18	3.32	4.38	5.17	5.50	5.43	4.64	3.56	2.09	1.13	0.84	
	RadStd	0.13	0.24	0.24	0.38	0.48	0.41	0.53	0.30	0.23	0.17	0.11	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page